

K.BALAJI

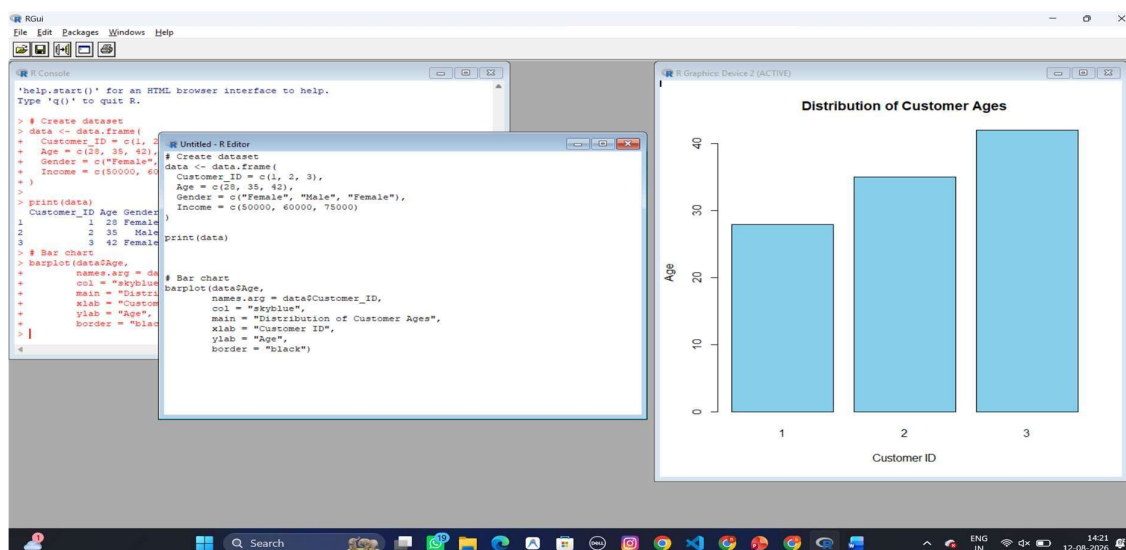
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DATA HANDLING AND VISUVALIZATION LAB 16-20

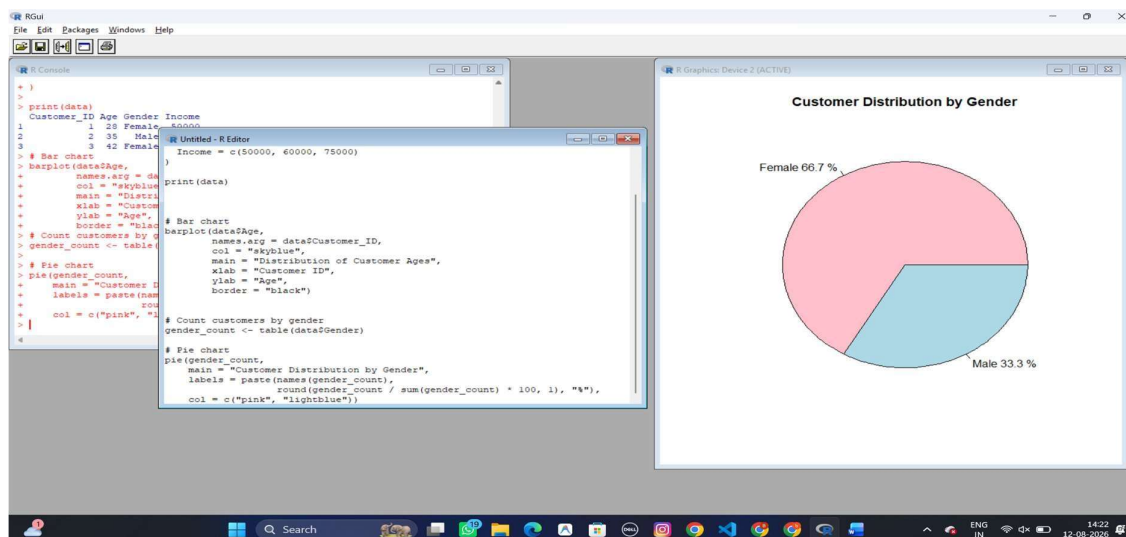
16) Customer Demographics Analysis

Dataset: Customer Demographics

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.



3. Build a table to show the demographic information for each customer. Label the table elements.

```

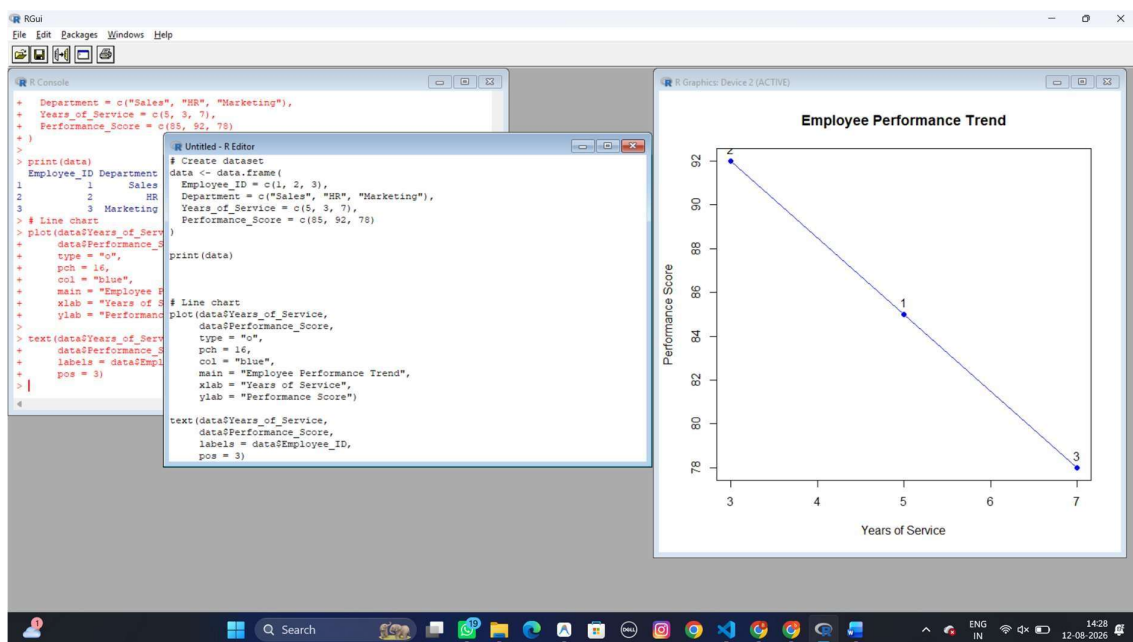
# Count customers by gender
gender_count <- table(data$Gender)
>
# Pie chart
pie(gender_count,
  main = "Customer Distribution by Gender",
  labels = paste(names(gender_count),
    round(gender_count / sum(gender_count) * 100, 1), "%"),
  col = c("pink", "lightblue"))
# Display table
print(data)
Customer_ID Age Gender Income
1 28 Female 50000
2 35 Male 60000
3 42 Female 75000
# Add labels
colnames(data) <- c("Customer ID", "Age", "Gender", "Income ($)")
print(data)
Customer ID Age Gender Income ($)
1 28 Female 50000
2 35 Male 60000
3 42 Female 75000

```

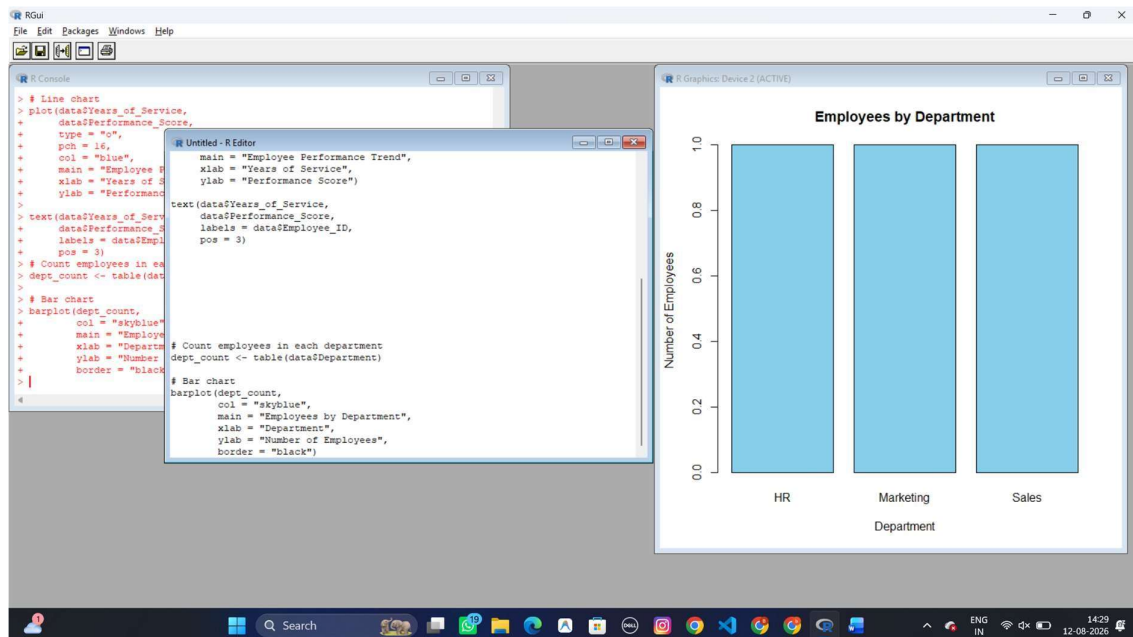
17: Employee Performance Analysis

Dataset: Employee Performance

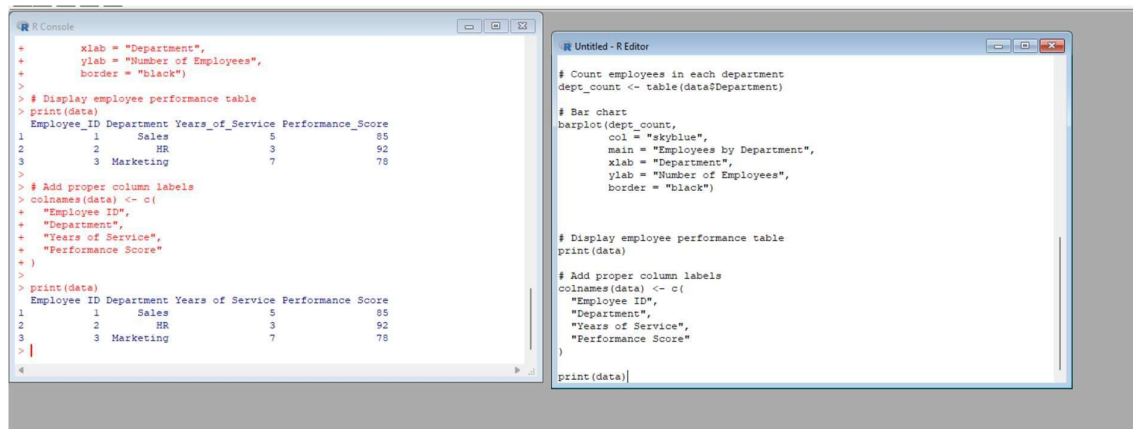
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.



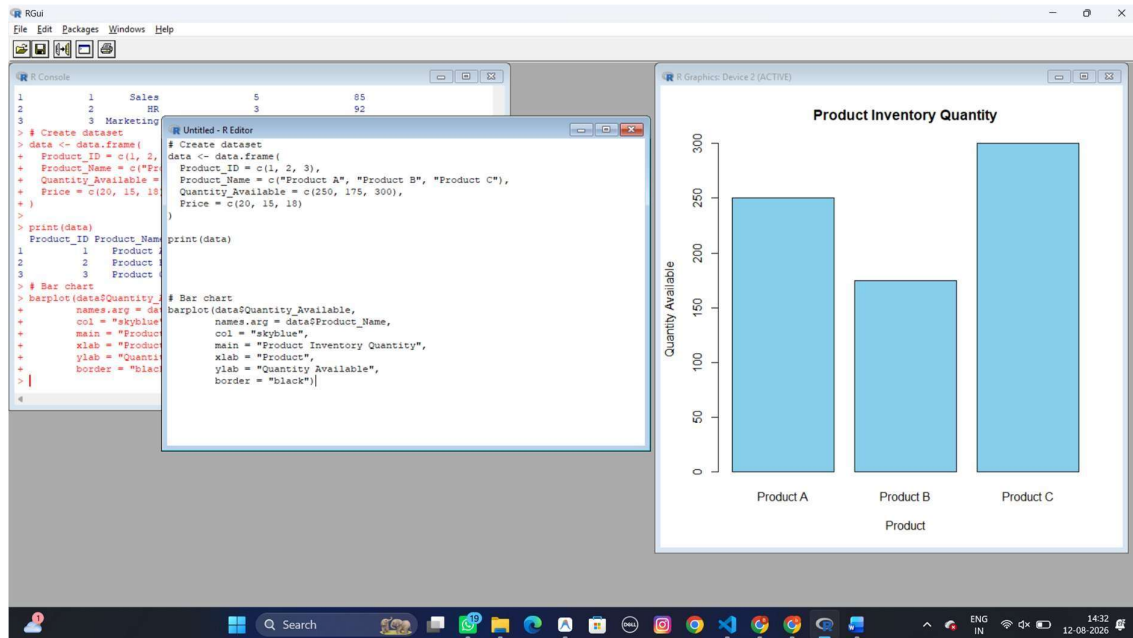
3. Build a table to display the performance data for each employee. Label the table elements.



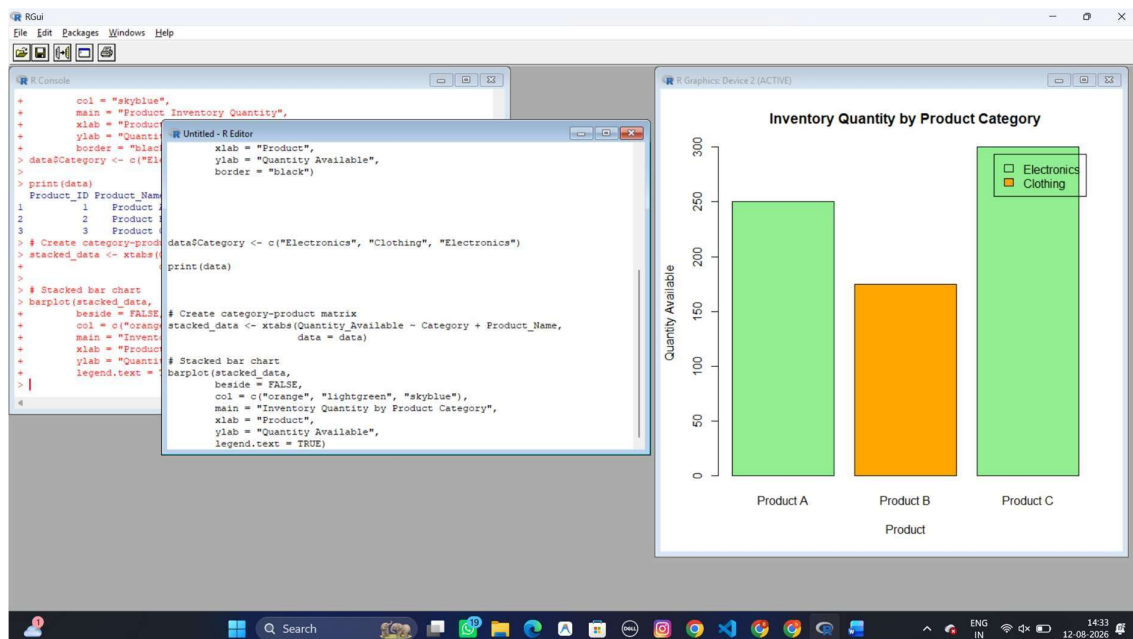
18: Product Inventory Management

Dataset: Product Inventory

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.



3. Build a table to show the inventory data for each product. Label the table elements.

```

# R Console
> # Stacked bar chart
> barplot(stacked_data,
+         beside = FALSE,
+         col = c("orange", "lightgreen", "skyblue"),
+         main = "Inventory Quantity by Product Category",
+         xlab = "Product",
+         ylab = "Quantity Available",
+         legend.text = TRUE)
> # Rename columns
> colnames(data) <- c(
+   "Product ID",
+   "Product Name",
+   "Quantity Available",
+   "Price ($)",
+   "Category"
+ )
> # Display table
> print(data)
  Product ID Product Name Quantity Available Price ($)  Category
1         1   Product A         250           20 Electronics
2         2   Product B         175           15   Clothing
3         3   Product C         300           18 Electronics

# R Editor
barplot(stacked_data,
        beside = FALSE,
        col = c("orange", "lightgreen", "skyblue"),
        main = "Inventory Quantity by Product Category",
        xlab = "Product",
        ylab = "Quantity Available",
        legend.text = TRUE)

# Rename columns
colnames(data) <- c(
  "Product ID",
  "Product Name",
  "Quantity Available",
  "Price ($)",
  "Category"
)

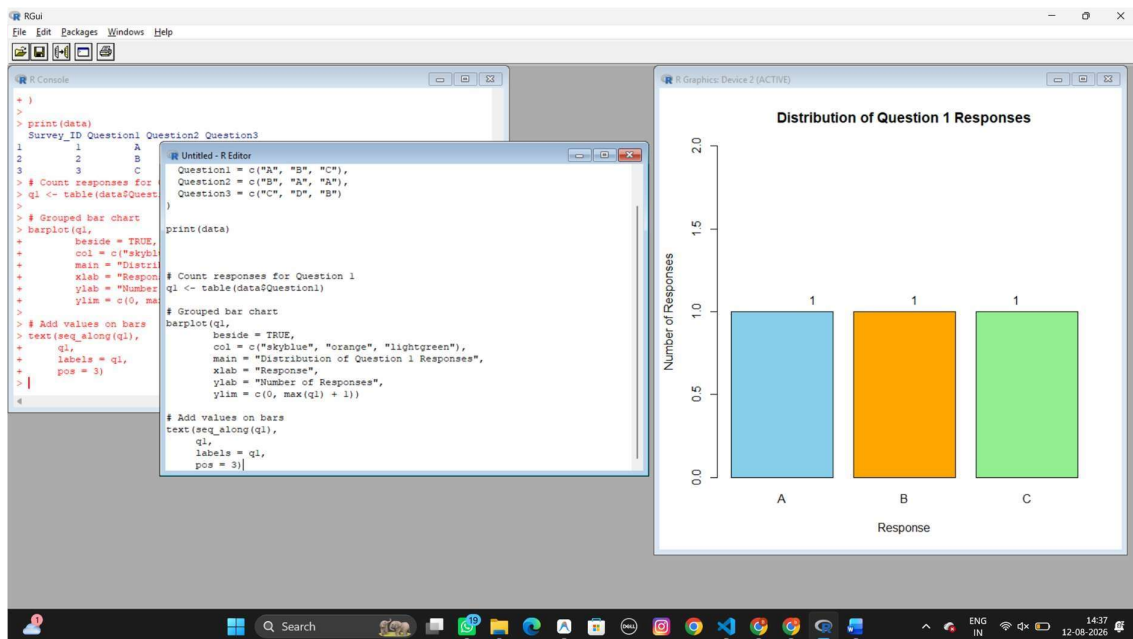
# Display table
print(data)

```

Set 19: Survey Responses Analysis

Dataset: Survey Responses

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```

R Console
> response_matrix <- matrix(
+   0,
+   nrow = length(responses),
+   ncol = 3
+ )
> 
> rownames(response_matrix) <- responses
> colnames(response_matrix) <- c(
+   "Question 1",
+   "Question 2",
+   "Question 3"
+ )
> 
> # Fill counts
> response_matrix[names(q1), "Question 1"] <- q1
> response_matrix[names(q2), "Question 2"] <- q2
> response_matrix[names(q3), "Question 3"] <- q3
> 
> print(response_matrix)
  Question 1 Question 2 Question 3
A          1          2          0
B          1          1          1
C          1          0          1
D          0          0          1
> 
> 
4

Untitled - R Editor
responses <- sort(unique(c(
  data$Question1,
  data$Question2,
  data$Question3
)))

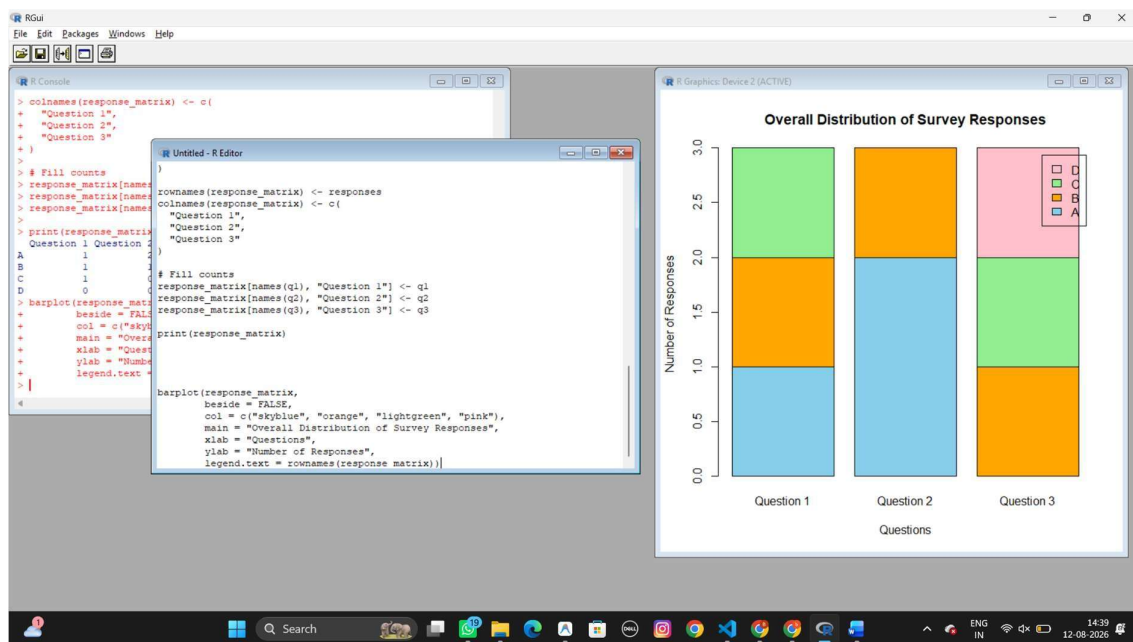
# Create matrix
response_matrix <- matrix(
  0,
  nrow = length(responses),
  ncol = 3
)

rownames(response_matrix) <- responses
colnames(response_matrix) <- c(
  "Question 1",
  "Question 2",
  "Question 3"
)

# Fill counts
response_matrix[names(q1), "Question 1"] <- q1
response_matrix[names(q2), "Question 2"] <- q2
response_matrix[names(q3), "Question 3"] <- q3

print(response_matrix)

```



3. Build a table to show the survey response data for each survey.
Label the table elements.

```

R Console
B          1          1          1
C          1          0          1
D          0          0          1
> barplot(response_matrix,
+   beside = FALSE,
+   col = c("skyblue", "orange", "lightgreen", "pink"),
+   main = "Overall Distribution of Survey Responses",
+   xlab = "Questions",
+   ylab = "Number of Responses",
+   legend.text = rownames(response_matrix))
> 
> # Rename columns
> colnames(data) <- c(
+   "Survey ID",
+   "Question 1",
+   "Question 2",
+   "Question 3"
+ )
> 
> # Display table
> print(data)
  Survey ID Question 1 Question 2 Question 3
1          1          A          B          C
2          2          B          A          D
3          3          C          A          B
> 
> 
4

Untitled - R Editor
barplot(response_matrix,
  beside = FALSE,
  col = c("skyblue", "orange", "lightgreen", "pink"),
  main = "Overall Distribution of Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses",
  legend.text = rownames(response_matrix))

# Rename columns
colnames(data) <- c(
  "Survey ID",
  "Question 1",
  "Question 2",
  "Question 3"
)

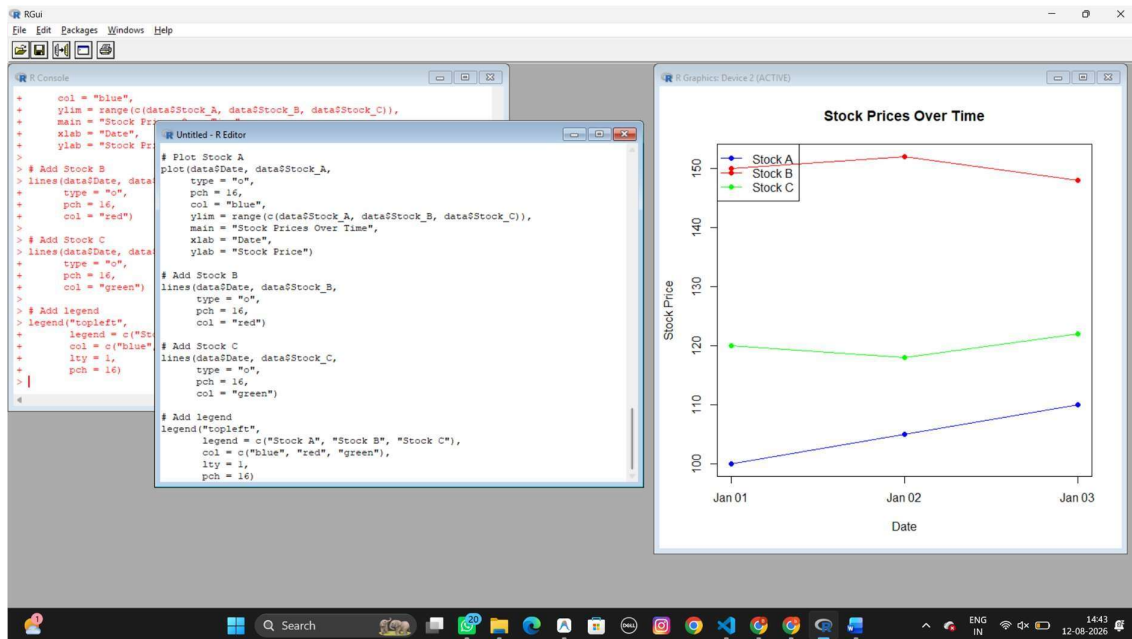
# Display table
print(data)

```

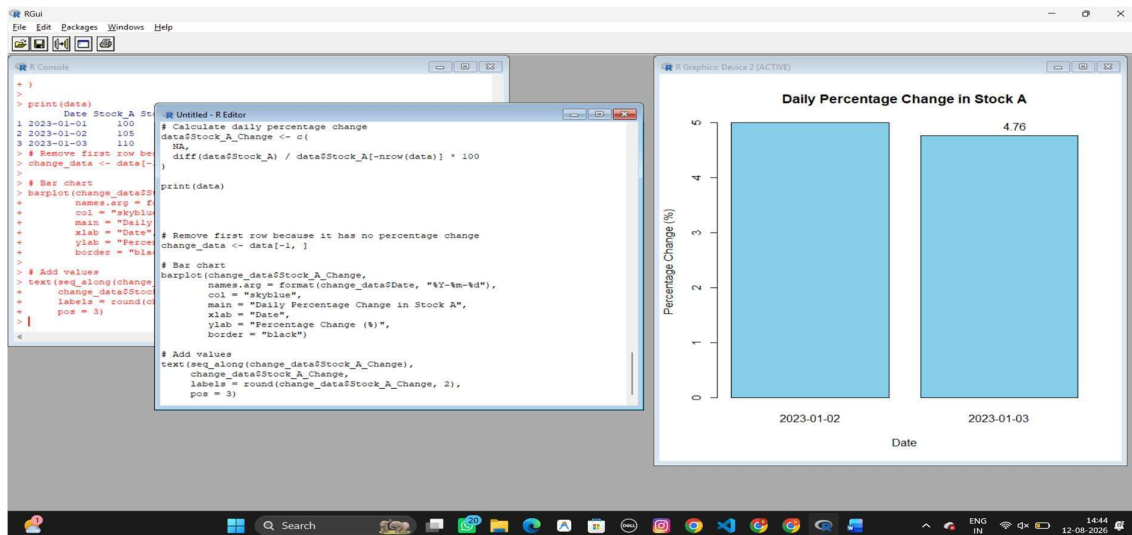

Set 20: Stock Analysis

Dataset: Stock Prices

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.



2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

R Console

```
3 2023-01-03      110      148      122      4.761905
> # Remove first row because it has no percentage change
> change_data <- data[-1, ]
>
> # Bar chart
> barplot(change_data$Stock_A_Change,
+         names.arg = format(change_data$Date, "%Y-%m-%d"),
+         col = "skyblue",
+         main = "Daily Percentage Change in Stock A",
+         xlab = "Date",
+         ylab = "Percentage Change (%)",
+         border = "black")
>
> # Add values
> text(seq_along(change_data$Stock_A_Change),
+      change_data$Stock_A_Change,
+      labels = round(change_data$Stock_A_Change, 2),
+      pos = 3)
> # Display stock price table
> print(data)
      Date Stock_A Stock_B Stock_C Stock_A_Change
1 2023-01-01      100      150      120           NA
2 2023-01-02      105      152      118      5.000000
3 2023-01-03      110      148      122      4.761905
>
```

Untitled - R Editor

```
)
print(data)

# Remove first row because it has no percentage change
change_data <- data[-1, ]

# Bar chart
barplot(change_data$Stock_A_Change,
        names.arg = format(change_data$Date, "%Y-%m-%d"),
        col = "skyblue",
        main = "Daily Percentage Change in Stock A",
        xlab = "Date",
        ylab = "Percentage Change (%)",
        border = "black")

# Add values
text(seq_along(change_data$Stock_A_Change),
     change_data$Stock_A_Change,
     labels = round(change_data$Stock_A_Change, 2),
     pos = 3)

# Display stock price table
print(data)
```